

Diabetes Risk Prediction

Predicting diabetes from 21 health indicators using supervised machine learning

GROUP

Ali Raza · 2540010
M. Ammar · 2540004
Taha Ali · 2540008

INSTRUCTOR

Sir Zaki

SEMESTER · DATE

Semester 2 · 2026

RESOURCES

GitHub: <https://github.com/muhammadammarff-design/diabetes-risk-prediction> · **Live demo:** <https://diabetesriskpredictions.streamlit.app/>

What we will cover today

- 01 · Problem statement
- 02 · Dataset overview
- 03 · Target & features
- 04 · Our model (Random Forest)
- 05 · Project workflow (block diagram)
- 06 · Data cleaning
- 07 · Exploratory data analysis
- 08 · Correlation analysis
- 09 · Train / test split
- 10 · Why machine learning?
- 11 · 4 candidate ML models
- 12 · Model comparison
- 13 · Random Forest — detailed results
- 14 · Top diabetes risk factors
- 15 · Streamlit web app
- 16 · Limitations
- 17 · Conclusion & results
- 18 · Q&A

The diabetes problem

- Diabetes affects 1 in 11 adults worldwide (~537 million people).
- Early detection drastically reduces complications and costs.
- Many people are unaware they have prediabetes or diabetes.
- Doctors need low-cost, fast screening tools.
- Patient survey data is cheap to collect — no lab tests required.

GLOBAL IMPACT

537M

adults living with diabetes

Source: IDF Diabetes Atlas 2021

Dataset overview

- Source: CDC Behavioral Risk Factor Surveillance System (BRFSS), 2015.
- Collected through annual phone interviews across the United States.
- Cleaned version published on Kaggle by Alex Teboul.
- Pre-balanced 50 / 50 between diabetic and non-diabetic patients.
- Each row = one patient · each column = one health indicator.

DATASET AT A GLANCE

Records	70,693 patients
Total Features	21 health indicators
Target	Diabetes_binary (0 / 1)
Class Balance	50 % / 50 %
Type	Tabular · structured
Task	Binary classification

Target and selected features

TARGET VARIABLE (y)

Diabetes_binary → 0 = No Diabetes · 1 = Diabetes / Prediabetes

SELECTED FEATURES (top 5 by model importance)

GenHlth (0.191) HighBP (0.166) BMI (0.149) Age (0.110) HighChol (0.079)

ALL 21 FEATURES (X) · ORGANISED IN 3 CATEGORIES

VITAL · CLINICAL (7)

- **HighBP** · high blood pressure
- **HighChol** · high cholesterol
- CholCheck · 5-yr check
- **BMI** · **Body Mass Index**
- Stroke · ever had one
- HeartDiseaseorAttack
- DiffWalk · difficulty walking

LIFESTYLE (7)

- Smoker · 100+ cigs
- PhysActivity · past 30 days
- Fruits · daily
- Veggies · daily
- HvyAlcoholConsump
- AnyHealthcare · coverage
- NoDocbcCost · couldn't see

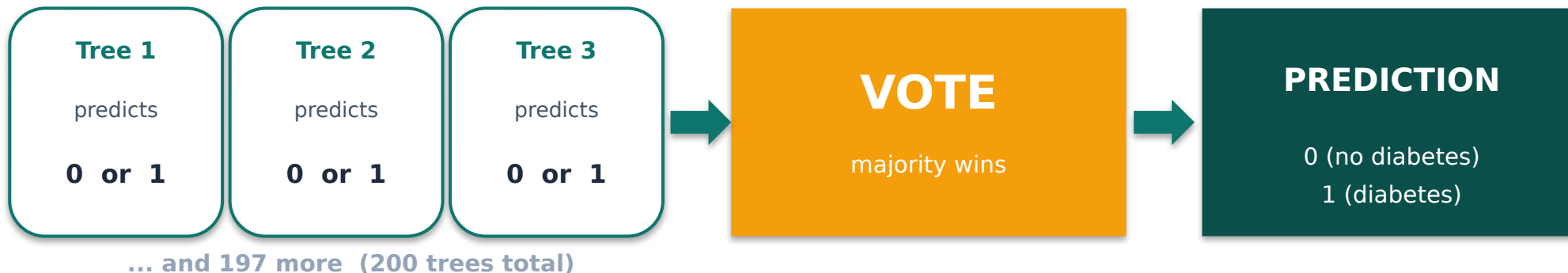
SELF-REPORTED · DEMO (7)

- **GenHlth** · **health (1-5)**
- MentHlth · bad-mood days
- PhysHlth · sick days
- Sex · 0 female / 1 male
- **Age** · **group (1-13)**
- Education · level (1-6)
- Income · bracket (1-8)

Random Forest

A team of 200 decision trees that vote on the answer.

Each tree looks at the patient data and predicts 0 or 1. The majority vote becomes the final prediction.



WHY: best F1 score across 4 models tested, and gives feature importance for free.

Block diagram: Input → Methodology → Output

INPUT

70,693 patient records · 21 health indicators · Diabetes_binary (target)

Source: BRFSS 2015 (CDC survey) — Kaggle clean version

METHODOLOGY

01 Data Cleaning

Drop duplicates



02 EDA & Correlation

Stats + heatmap



03 Train / Test Split

80/20 stratified



04 Train Random Forest

200 trees

OUTPUT

ACCURACY

74.36%

F1 SCORE

0.759

RECALL

0.794

Data cleaning — what we did

- Checked for missing values → found 0 ✓
- Removed duplicate rows → 1 636 dropped
- Final shape: 69 057 rows × 22 columns
- Converted 21 columns to int (everything except BMI which is float)
- Verified class balance still ~50 / 50

BEFORE → AFTER

Rows

70 693 → 69 057

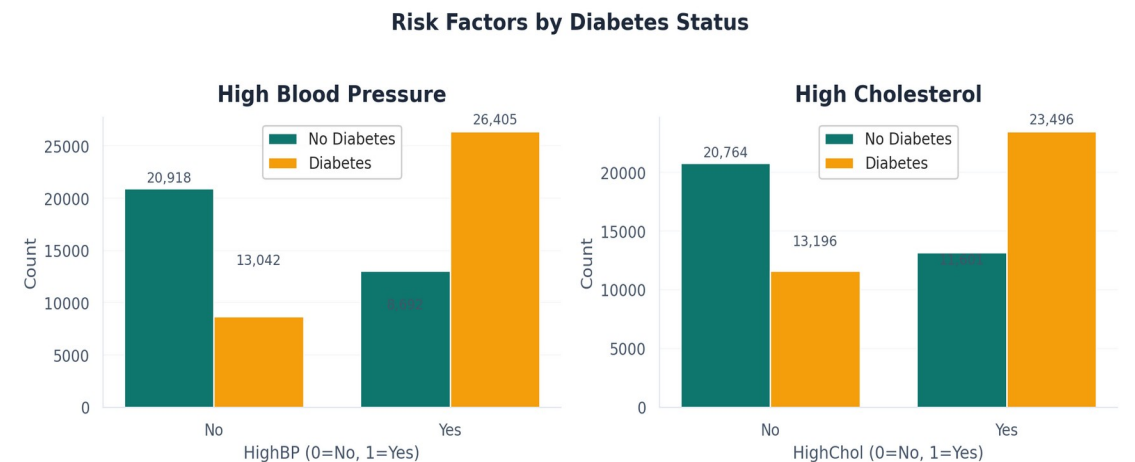
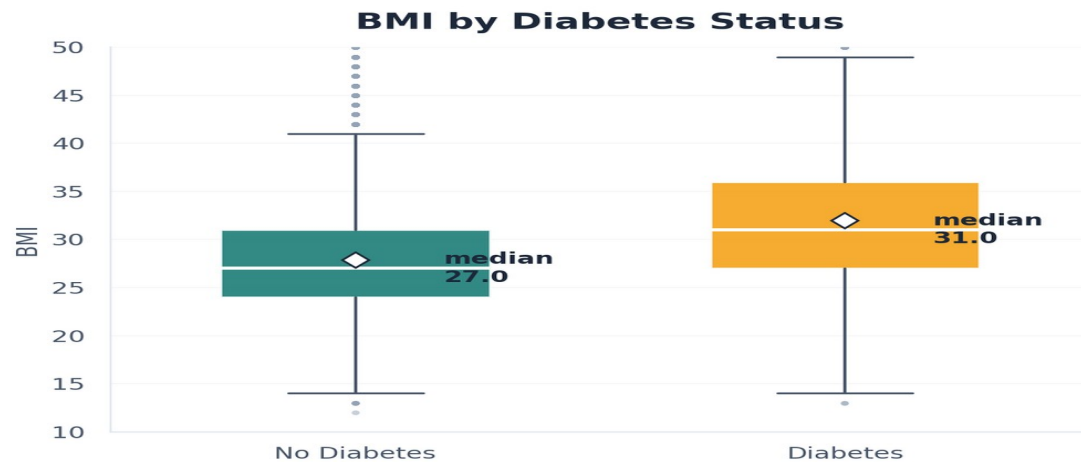
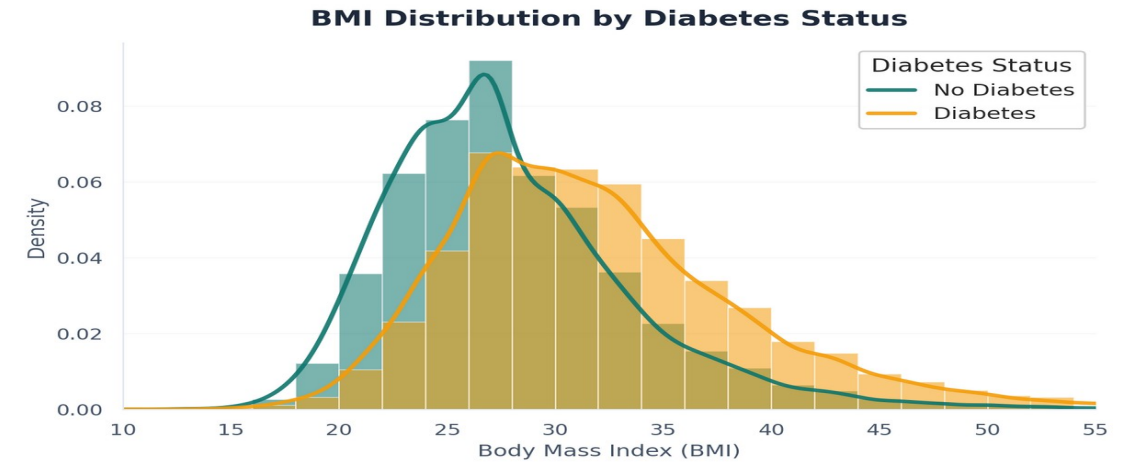
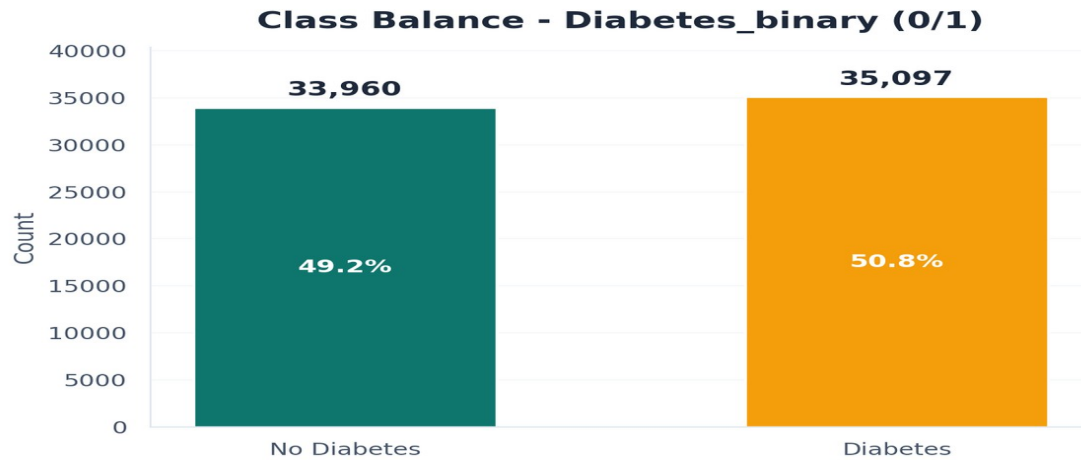
Duplicates

1 636 → 0

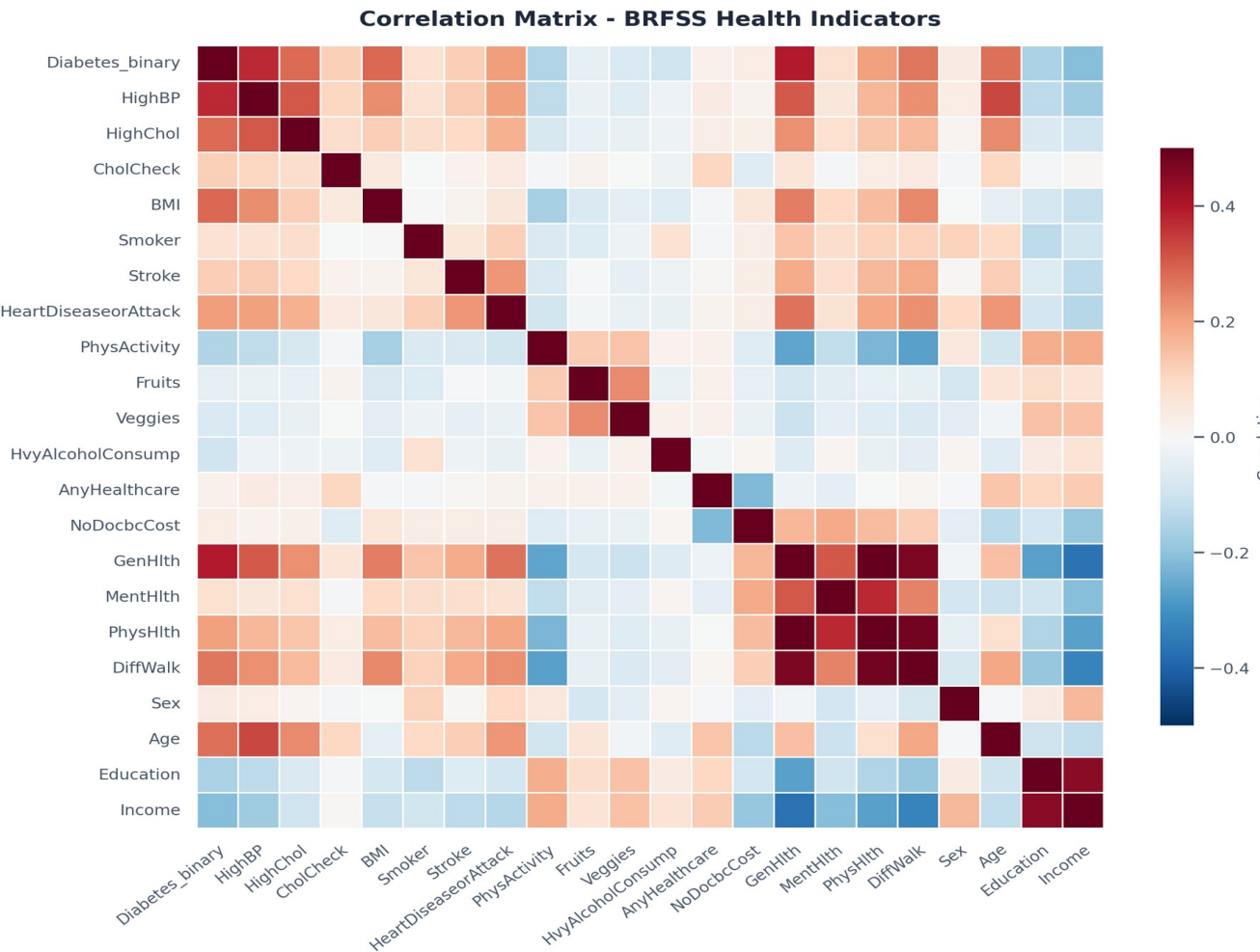
Missing values

0 → 0 ✓ already clean

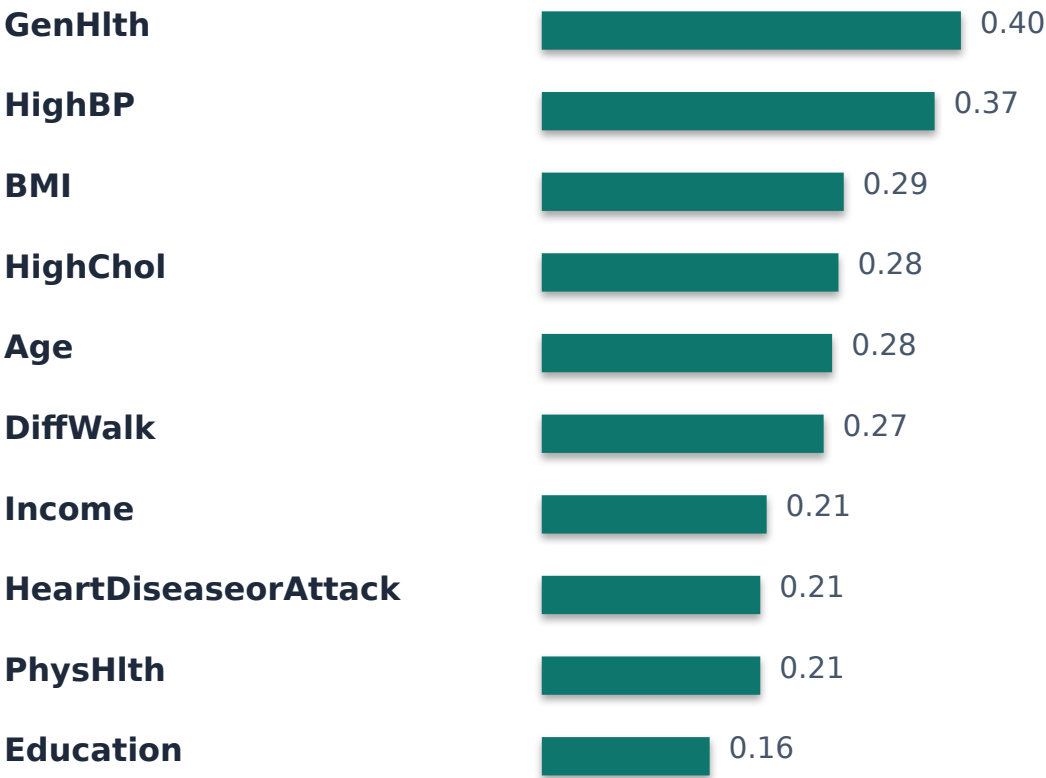
Exploratory data analysis



Which features matter most?



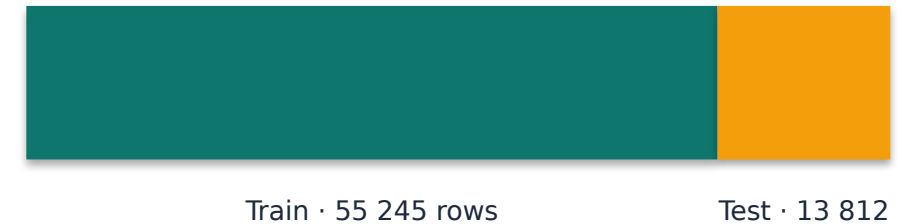
TOP 10 CORRELATED WITH DIABETES



Train / test split — done before everything else

- 80 % training data · 20 % test data
- Stratified by target → keeps 50 / 50 balance in both splits
- `random_state = 42` → reproducible results
- Split BEFORE any preprocessing → prevents data leakage

DATA SPLIT



WHY SPLIT FIRST?

If we scaled, imputed, or sampled before splitting, information from the test set would leak into training. Accuracy would look great in our notebook but fail on new data.

Why supervised machine learning?

LEARNING TYPE

Supervised

The target column is labeled (0 / 1) — the model learns from known examples.

TASK

Classification

We predict a discrete class, not a number.
Two classes → binary classification.

OUTPUT

Probability + label

Each prediction includes a confidence score, useful for risk-screening contexts.

4 supervised classifiers tested

LOGISTIC REGRESSION

Fits a probability line between the two classes

PROS Simple · interpretable

CONS Less accurate on non-linear patterns

DECISION TREE

Asks yes/no questions to split the data

PROS Easy to visualise

CONS Prone to overfitting

K-NEAREST NEIGHBORS

Predicts class by voting among nearest neighbours

PROS Intuitive

CONS Slow on large data · needs scaling

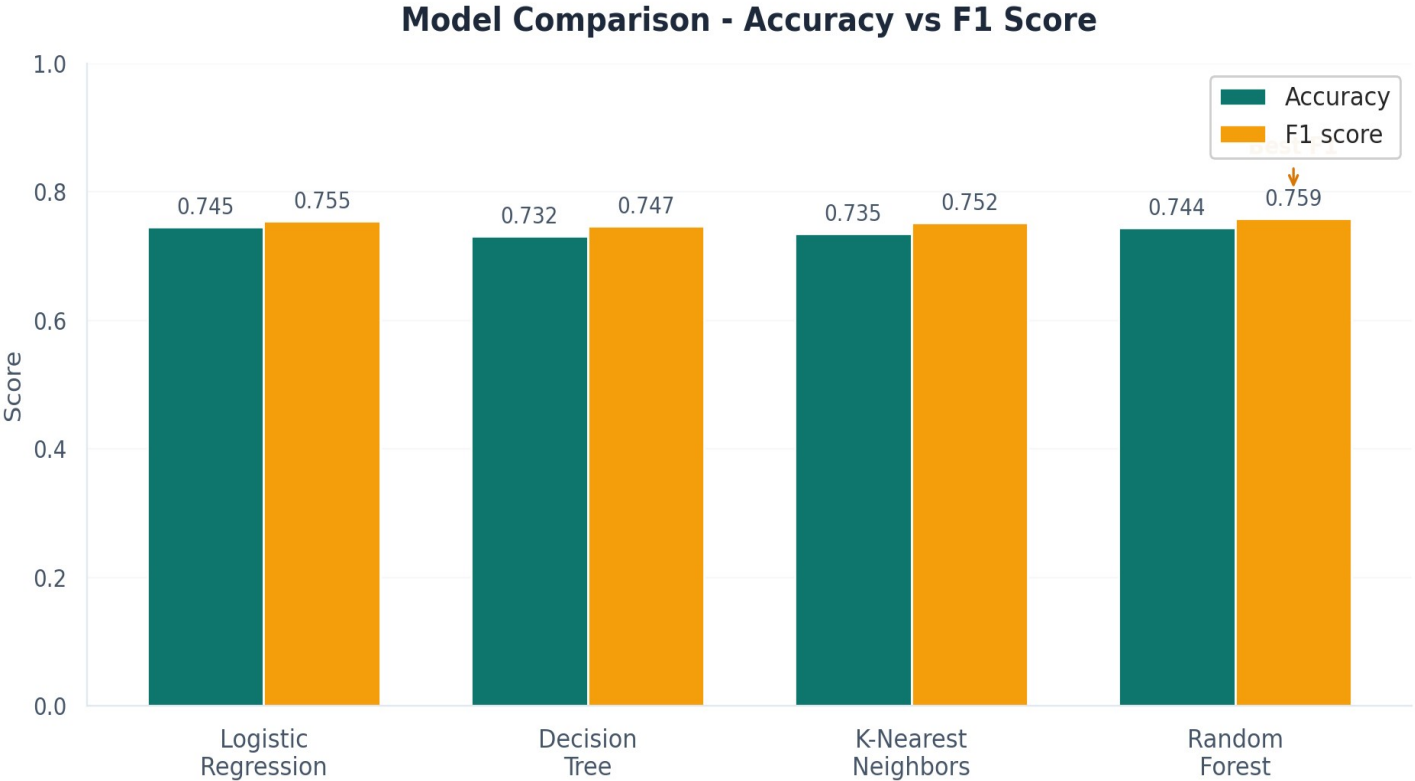
RANDOM FOREST

Combines many decision trees, votes the answer ✓ SELECTED

PROS Robust · feature imp.

CONS Slower to train

Model comparison — real numbers



FULL METRICS

Model	Acc	F1
Logistic Regression	0.745	0.755
Decision Tree	0.732	0.747
K-Nearest Neighbors	0.735	0.752
Random Forest	0.744	0.759

WINNER

Random Forest

Best F1 (0.759) and built-in feature importance.

Random Forest — detailed results

Confusion Matrix - Random Forest

Actual label	Predicted label	
	No Diabetes	Diabetes
No Diabetes	4696	2096
Diabetes	1446	5574

ACCURACY

74.36%

correct predictions

F1 SCORE

0.759

best of 4 models

PRECISION

0.727

predicted diabetics correct

RECALL

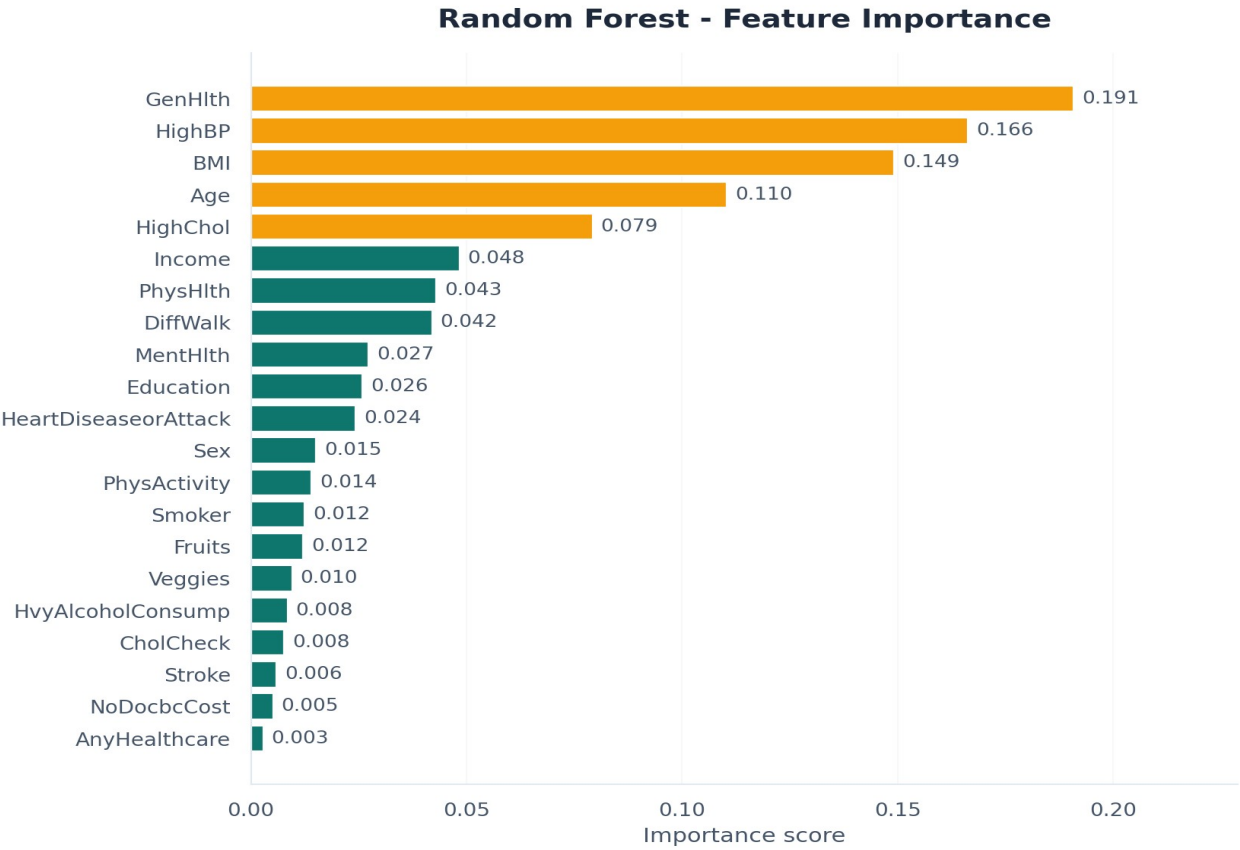
0.794

actual diabetics we caught

HYPERPARAMETERS

n_estimators=200 · max_depth=15 · min_samples_leaf=5 · random_state=42

Top diabetes risk factors



Top 5 highlighted in amber: GenHlth (0.191), HighBP (0.166), BMI (0.149), Age (0.110), HighChol (0.079)

TOP 5 RISK FACTORS LEARNED

- 1. **GenHlth**
importance = 0.191
- 2. **HighBP**
importance = 0.166
- 3. **BMI**
importance = 0.149
- 4. **Age**
importance = 0.110
- 5. **HighChol**
importance = 0.079

Streamlit web application

- Interactive form built with Streamlit (Python web framework).
- User enters 21 health indicators in plain English.
- Trained model loaded once and cached for speed.
- Output: risk label + probability + confidence bar.
- Code on GitHub · deployed free on Streamlit Cloud.

DIABETES RISK PREDICTION

BMI	27
High BP	Yes
High Chol	No
General Hlth	3 - Good
Age group	50-54

PREDICT

HIGH RISK
probability 68 %

Results · Limitations · Q&A

ACCURACY	F1 SCORE	PRECISION	RECALL
74.36% on test set	0.759 best of 4 models	0.727 of predicted diabetics	0.794 of real diabetics caught

CONCLUSION

- Random Forest → 74.4% accuracy · 0.759 F1 score.
- 5 dominant risk factors identified, aligned with medical literature.
- Working web app + reproducible notebook, GitHub-ready.

LIMITATIONS

- Self-reported survey data is noisy.
- No lab values (no HbA1c or glucose).
- Balanced data $\neq$ real-world prevalence (~14%).

LIKELY QUESTIONS

Why Random Forest over Logistic Reg?

Best F1 + feature importance for free.

Why only 74% accuracy?

Self-reported data caps it; matches published research.

Why drop duplicates?

Prevents test-set inflation (data leakage).

Supervised or unsupervised?

Supervised — target is labeled 0 / 1.

How would you improve it?

Add lab values · try XGBoost · tune hyperparameters.

Thank you · Questions?

Code · github.com/muhammadammarff-design/diabetes-risk-prediction · **Live demo** · diabetesriskpredictions.streamlit.app